data," says Sanjay Sarin, head of country operations for India at FIND or the Foundation for Innovative New Diagnostics. "So, if I want AI to analyse a TB (tuberculosis) patient's reports, I need to feed it a range of previous reports of TB patients from different regions and populations. The more varied the data, the more robust and accurate the analysis and predictions." (TB is the only communicable disease to be included in predictive health programmes, though AI diagnostics can help only with early, not prior, detection.)

AI diagnostics is also deploying deep-learning algorithms to analyse radiology material like X-rays, ECGs, CT scans and MRIs to identify potential existing abnormalities as well as predict future ones based on the current state of the organ. FIND, for instance, has been working with different organisations in India for AI-imaging products to diagnose TB. "Imaging is one area where AI has worked really well to give quick and accurate results," says Sarin. "You'll find AI-imaging tools for TB even in district hospitals today. The cost is only a one-time investment in hardware. In fact, in some cases, even analog machines can be converted to digital ones."

AI has impacted imaging diagnostics so markedly that some products can interpret and analyse ECG findings within a few minutes of being fed a report. Tricog, a Bengaluru-based company, for example, has a product called InstaECG that can analyse an ECG report in under 10 minutes while another product, InstaEcho, assists doctors in the ECG diagnosis to ensure nothing has been missed. More than 5,000 catheterisation labs, hospitals, clinics and diagnostic centres are using these to help diagnose and manage patients with critical cardiac illnesses. Tricog is now working with the global biopharma firm AstraZeneca on Project Heartbeat, a tool that can interpret ECGs and guide patients to the nearest cath lab.

AI's most significant contribution has been in processing genetic information, especially for cardiac disease, metabolic disorders and cancer, wherein machines are trained to identify certain genetic red flags. For around Rs 14,000 each, the Delhi- and Mohali-based DNA Forensics Laboratory offers tests that can predict hypertension, autoimmune illnesses, skin diseases, eye ailments, sleep disorders and even thrombosis risk by examining mutations commonly associated with the disease. MedGenome, on the other hand, offers over 1,300 types of genetic tests in India. "Technology is allowing us to process a larger volume of reports and compare them with a wider database of existing genetic information," says Dr Ramesh Menon, associate director, Genomic Medicine & Personal Genomics Divisions, Bioinformatics Department, MedGenome. "This way, you get more accurate and faster results with DNA testing."

"The presence of certain genes means you have a statistically higher chance of getting a certain disease," says Dr Bharat Aggarwal, principal director of radiology services at Max Hospital Saket, Delhi. "It is not definitive, but there is a risk. You then come in for more screenings, so that the disease can be detected as early as possible, and you can make lifestyle changes to delay its onset as well as severity. This improves the longevity and quality of your life."

THE PITFALLS OF PREDICTIVE HEALTHCARE

Some caveats are in order. The saga of American biotech entrepreneur Elizabeth Holmes, the founder and chief executive of Theranos, is only the latest cautionary tale. It was the vision of predicting disease before it made its actual appearance that she sold to investors in her blood testing technology, before it was discovered that it was not as revolutionary as she made it out to be. Similarly, IBM's Watson supercomputer, which a decade ago was hailed as a game-changer in cancer diagnostics and treatment, is now under the scanner for having provided unsafe recommendations.

"The FDA clearance for the Apple Watch ECG application includes validation data from a trial of around 600 people only," says Dr Dastidar. "What's needed are large-scale clinical studies for such applications, like for drugs, vaccines or traditional diagnostic devices." For an AI-powered machine to predict long-term risks, it needs

"You need to train an algorithm on data...**feed it a range of previous reports from different regions and populations.** The more varied the data, the more robust and accurate the analysis and predictions"

–SANJAY SARIN
Head of country operations for India, FIND (Foundation for Innovative New Diagnostics)

